9TH GRADE CHEMISTRY · NON-AP · ALA VISTANCIA · MR/MS SHARP

Foundations of Chemistry

FINAL EXAM · 50 QUESTIONS · 100 POINTS

SECTION A · VOCABULARY · 17 pts SECTION B · MULTIPLE CHOICE · 20 pts

SECTION C · FILL-IN-BLANK · 13 pts

SECTION A · 17 POINTS

Vocabulary

Read each definition. Select the correct chemistry term from the dropdown. · 1 point each

Q1 · VOCABULARY

The smallest particle of an element that still retains that element's chemical properties is a(n) — select term — .

Q2 · VOCABULARY

A pure substance composed of only one type of atom — which cannot be broken down by ordinary chemical means — is called a(n) — select term — .

Q3 · VOCABULARY

A pure substance formed when two or more elements chemically bond together in a fixed ratio is called a(n) — select term — .

Q4 · VOCABULARY

A combination of two or more substances that are NOT chemically combined — where each substance keeps its own properties — is called a – select term –.

Q5 · VOCABULARY

A positively charged (+) subatomic particle found in the nucleus of an atom is called a – select term –.

Q6 · VOCABULARY

A neutral (no charge) subatomic particle found in the nucleus of an atom is called a – select term –.

Q7 · VOCABULARY

A negatively charged (−) subatomic particle that moves in orbitals around the nucleus is called a(n) – select term –.

Q8 · VOCABULARY

The number of protons in an atom's nucleus — which is unique to every element and never changes — is called the – select term –.

Q9 · VOCABULARY

The total count of protons plus neutrons in an atom's nucleus is called the – select term –.

Q10 · VOCABULARY

Atoms of the same element that have the same atomic number but different numbers of neutrons are called – select term –.

Q11 · VOCABULARY

An atom or group of atoms that carries an electric charge because it has gained or lost electrons is called a(n) — select term — .

Q12 · VOCABULARY

The electrons in the outermost energy level of an atom that participate in chemical bonding are called — select term — electrons.

Q13 · VOCABULARY

A chemical bond formed when two atoms SHARE one or more pairs of electrons is called a — select term — bond.

Q14 · VOCABULARY

A chemical bond formed by the complete TRANSFER of electrons from a metal to a nonmetal is called a(n) — select term — bond.

Q15 · VOCABULARY

The SI unit for amount of substance in chemistry — equal to 6.022×10^{23} representative particles — is the — select term — .

Q16 · VOCABULARY

A substance that is present at the START of a chemical reaction and is consumed (used up) during the reaction is called a — select term — .

Q17 · VOCABULARY

A new substance that is FORMED as a result of a chemical reaction — written on the right side of the equation — is called a — select term — .

SECTION B · 20 POINTS

Multiple Choice

Select the single best answer from the six options. Calculation models are available for math-based questions. · 1 point each

L01 · MATTER

Q18 · L01 · MATTER

Which of the following is an example of a **physical change**?

A. Iron rusting after exposure to rain

B. Natural gas burning in a stove flame

C. Milk curdling when it spoils

D. Ice melting into liquid water

E. Wood rotting over several years

F. Baking soda fizzing when mixed with vinegar

Q19 · L01 · MATTER

Salt water (NaCl dissolved in H_2O) is best classified as a...

A. Pure element

B. Pure compound

C. Homogeneous mixture

D. Heterogeneous mixture

E. Suspension

F. Single-element substance

L01 · MEASUREMENT

Q20 · L01 · MEASUREMENT

A metal sample has a mass of 45.0 g and a volume of 5.0 cm^3. What is its **density**?

A. 0.11 g/cm^3

B. 225 g/cm^3

C. 50.0 g/cm^3

D. 40.0 g/cm^3

E. 9.0 g/cm^3

F. 4.5 g/cm^3

L02 · ATOMIC STRUCTURE

Q21 · L02 · ATOMIC STRUCTURE

Where is the majority of an atom's **mass** located?

A. In the electron cloud surrounding the nucleus

B. Evenly distributed throughout the atom

C. In the outermost electron shells

D. In the valence electrons only

E. In the nucleus

F. In the space between the nucleus and electrons

Q22 · L02 · ATOMIC STRUCTURE

An atom has **17 protons** and **18 neutrons**. What is its **mass number**?

A. 17

B. 18

C. 1

D. 35

E. 34

F. 51

Q23 · L02 · ATOMIC STRUCTURE

Which subatomic particle determines the **identity** (which element) of an atom?

A. Neutron

B. Electron

C. Proton

D. Both protons and neutrons equally

E. Ion

F. Valence electron

L03 · PERIODIC TABLE

Q24 · L03 · PERIODIC TABLE

Elements in the same **group** (vertical column) on the periodic table share similar...

A. Atomic masses

B. Numbers of neutrons

C. Mass numbers

D. Physical states at room temperature

E. Numbers of electron shells

F. Chemical properties

Q25 · L03 · PERIODIC TABLE

Which of the following elements is classified as a **noble gas**?

A. Nitrogen (N)

B. Fluorine (F)

C. Chlorine (Cl)

D. Oxygen (O)

E. Sodium (Na)

F. Argon (Ar)

Q26 · L03 · PERIODIC TABLE

Moving from left to right across Period 3 (Na → Ar), which property **steadily increases**?

A. Atomic radius

B. Metallic character

C. Number of electron shells

D. Reactivity with water

E. Atomic number

F. Number of neutrons in all atoms

L04 · CHEMICAL BONDING

Q27 · L04 · CHEMICAL BONDING

Which type of chemical bond involves the complete **transfer** of electrons from one atom to another?

A. Covalent bond

B. Metallic bond

C. Polar covalent bond

D. Hydrogen bond

E. Van der Waals force

F. Ionic bond

Q28 · L04 · CHEMICAL BONDING

How many **valence electrons** does a carbon atom (Group 14) have?

A. 1

B. 2

C. 3

D. 6

E. 4

F. 14

Q29 · L04 · CHEMICAL BONDING

A water molecule (H_2O) has two lone pairs on oxygen. What **molecular shape** does this produce?

A. Linear

B. Trigonal planar

C. Bent (V-shaped)

D. Tetrahedral

E. Octahedral

F. T-shaped

L05 · CHEMICAL REACTIONS

Q30 · L05 · CHEMICAL REACTIONS

According to the **Law of Conservation of Mass**, what is always conserved in a chemical reaction?

A. Number of molecules only

B. Volume of all substances

C. Temperature of the system

D. Number of moles only

E. Color and physical state

F. Total mass and total number of each type of atom

Q31 · L05 · CHEMICAL REACTIONS

Which equation represents a **decomposition** reaction?

A. $2Na + Cl_2 \rightarrow 2NaCl$

B. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

C. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

D. $HCl + NaOH \rightarrow NaCl + H_2O$

E. $2H_2 + O_2 \rightarrow 2H_2O$

F. $2H_2O_2 \rightarrow 2H_2O + O_2$

Q32 · L05 · CHEMICAL REACTIONS

What do **coefficients** in a balanced chemical equation represent?

A. The charge of each ion

B. The number of atoms within one molecule

C. The subscripts of each element

D. The temperature of the reaction

E. The number of moles (or molecules) of each substance

F. The oxidation states of atoms

L06 · STOICHIOMETRY

Q33 · L06 · STOICHIOMETRY

Avogadro's number (6.022×10^{23}) represents the number of particles in exactly...

A. 1 gram of any substance

B. 1 liter of any gas at STP

C. 1 molecule of any compound

D. 1 mole of any substance

E. 1 atomic mass unit

F. 1 formula unit only

Q34 · L06 · STOICHIOMETRY

What is the **molar mass** of water, H_2O? (H = 1.0 g/mol, O = 16.0 g/mol)

A. 2.0 g/mol

B. 10.0 g/mol

C. 16.0 g/mol

D. 32.0 g/mol

E. 20.0 g/mol

F. 18.0 g/mol

Q35 · L06 · STOICHIOMETRY

How many **moles** are in 36.0 g of water? (Molar mass of H_2O = 18.0 g/mol)

A. 0.5 mol

B. 36.0 mol

C. 18.0 mol

D. 4.0 mol

E. 1.0 mol

F. 2.0 mol

L07 · STATES & SOLUTIONS

Q36 · L07 · STATES & SOLUTIONS

Which state of matter has a **definite volume** but **no definite shape**?

A. Solid

B. Gas

C. Plasma

D. Suspension

E. Liquid

F. Colloid

Q37 · L07 · STATES & SOLUTIONS

As temperature increases, what generally happens to the solubility of a **solid solute** in water?

A. Decreases significantly

B. Stays exactly the same

C. First increases, then decreases sharply

D. Drops to zero

E. Doubles instantly

F. Increases

SECTION C · 13 POINTS

Fill-in-the-Blank

Select the word or value that correctly completes each sentence. · 1 point each

F01 · L02 · ATOMIC STRUCTURE · Q38

The number of — select — in an atom's nucleus determines which element it is.

F02 · L02 · ATOMIC STRUCTURE · Q39

An atom that has gained or lost electrons becomes a(n) — select —.

F03 · L02 · ATOMIC STRUCTURE · Q40

Atoms of the same element with different numbers of neutrons are called — select —.

F04 · L03–L04 · BONDING · Q41

A [– select –] bond forms when two nonmetal atoms SHARE a pair of electrons.

F05 · L03 · PERIODIC TABLE · Q42

Elements in the same vertical column (same [– select –]) have the same number of valence electrons.

F06 · L05 · REACTIONS · Q43

In $2H_2 + O_2 \rightarrow 2H_2O$, the H_2 and O_2 on the LEFT side are called the [– select –].

F07 · L05 · REACTIONS · Q44

When balancing equations, we adjust the [– select –] in front of formulas — never the subscripts.

F08 · L01 · MATTER · Q45

Salt dissolving in water is a [– select –] change because no new substance is formed.

F09 · L02 · ATOMIC STRUCTURE · Q46

The dense central region of an atom that contains protons and neutrons is called the [– select –].

F10 · L06 · STOICHIOMETRY · Q47

The SI unit for amount of substance — equal to 6.022×10^{23} particles — is the [– select –].

F11 · L06 · STOICHIOMETRY · Q48

The molar mass of CO_2 is [– select –] g/mol. (C = 12 g/mol, O = 16 g/mol)

F12 · L07 · SOLUTIONS · Q49

A solution in which water is the solvent is called a(n) – select – solution.

F13 · L03 · PERIODIC TABLE · Q50

Elements that are shiny, malleable, ductile, and good electrical conductors are classified as – select –.